

Part II interacting Weyl reduction

The exact scalar convergence criterion and the excluded quartic LR citation

Lamport-form proof and proof-obligation ledger

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Reduction completed; interacting convergence remains open. To obtain a full fixed-coarse weak-star scaling limit, it is sufficient to prove convergence of the single scalar characteristic functional

$$\chi_{N,M}(\xi) := \omega_N(\alpha_M^N(W_M(\xi)))$$

for every fixed M and every coarse one-particle vector ξ . Positivity, boundedness, extension from Weyl polynomials, projective consistency, and continuum identification are then automatic by the proved steps below.

This static reduction requires neither an interacting Lieb–Robinson bound nor a uniform spectral gap. Those inputs are needed only for a quantitative spatial-cutoff or dynamical theorem.

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1 Characteristic-functional criterion

Let $(\mathcal{W}_N, \alpha_M^N)$ be the Weyl C^* -inductive system fixed in the model sheet. Let

$$\mathcal{D}_M := \text{span}\{W_M(\xi) : \xi \in h_{M,L}\}. \quad (1.1)$$

By the definition of the Weyl C^* -algebra,

$$\overline{\mathcal{D}_M}^{\|\cdot\|} = \mathcal{W}_M. \quad (1.2)$$

Theorem 1.1 (full-sequence Weyl criterion). *For every N , let $\omega_N \in S(\mathcal{W}_N)$. Assume that, for every fixed M and every $\xi \in h_{M,L}$, the limit*

$$\chi_M(\xi) := \lim_{N \rightarrow \infty, N \geq M} \omega_N(\alpha_M^N(W_M(\xi))) \quad (1.3)$$

exists. Then:

1. *there is a unique state $\rho_M \in S(\mathcal{W}_M)$ satisfying $\rho_M(W_M(\xi)) = \chi_M(\xi)$;*
2. *for every $A \in \mathcal{W}_M$,*

$$\lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(A)) = \rho_M(A); \quad (1.4)$$

3. *the family is projectively consistent,*

$$\rho_M = \rho_{M'} \circ \alpha_M^{M'}, \quad M \leq M'; \quad (1.5)$$

4. *there is a unique state $\Omega_\infty \in S(\mathcal{W}_\infty)$ with*

$$\Omega_\infty \circ \alpha_M^\infty = \rho_M. \quad (1.6)$$

2 Proof of Theorem 1.1

⟨1⟩ 1 Define the limit on finite Weyl polynomials.

For

$$D = \sum_{j=1}^s c_j W_M(\xi_j) \in \mathcal{D}_M, \quad (2.1)$$

set

$$\rho_M^{(0)}(D) := \sum_{j=1}^s c_j \chi_M(\xi_j). \quad (2.2)$$

This is independent of the representation (2.1). Indeed, if $D = 0$ in $\mathcal{W}_M$, then $\alpha_M^N(D) = 0$ for every N , and hence

$$\sum_j c_j \chi_M(\xi_j) = \lim_N \omega_N(\alpha_M^N(D)) = 0. \quad (2.3)$$

Linearity is immediate.

⟨1⟩ 2 Prove boundedness and positivity before extending.

For $D \in \mathcal{D}_M$, (1.3) and linearity of the finite sum give

$$\rho_M^{(0)}(D) = \lim_N \omega_N(\alpha_M^N(D)). \quad (2.4)$$

Because α_M^N is injective and therefore isometric,

$$|\rho_M^{(0)}(D)| \leq \limsup_N \|\alpha_M^N(D)\| = \|D\|. \quad (2.5)$$

If $D \in \mathcal{D}_M$, then $D^*D \in \mathcal{D}_M$, and

$$\rho_M^{(0)}(D^*D) = \lim_N \omega_N(\alpha_M^N(D)^* \alpha_M^N(D)) \geq 0. \quad (2.6)$$

Also $\rho_M^{(0)}(I) = 1$.

⟨1⟩ 3 Extend to the coarse C^* -algebra.

Equations (1.2) and (2.5) give a unique bounded extension $\rho_M : \mathcal{W}_M \rightarrow \mathbb{C}$. Positivity persists under norm limits: if $A \in \mathcal{W}_M$ and $D_j \rightarrow A$, then $D_j^* D_j \rightarrow A^* A$, so

$$\rho_M(A^* A) = \lim_j \rho_M^{(0)}(D_j^* D_j) \geq 0. \quad (2.7)$$

Together with $\rho_M(I) = 1$, this makes ρ_M a state.

⟨1⟩ 4 Upgrade convergence from Weyl polynomials to every coarse observable.

Fix $A \in \mathcal{W}_M$ and $\delta > 0$. By (1.2), choose $D \in \mathcal{D}_M$ with $\|A - D\| < \delta$. Then

$$|\omega_N(\alpha_M^N(A)) - \rho_M(A)| \leq |\omega_N(\alpha_M^N(A - D))| \quad (1)$$

$$+ |\omega_N(\alpha_M^N(D)) - \rho_M(D)| + |\rho_M(D - A)| \quad (2)$$

$$\leq 2\delta + |\omega_N(\alpha_M^N(D)) - \rho_M(D)|. \quad (2.8)$$

The last term tends to zero by (2.4). Therefore

$$\limsup_N |\omega_N(\alpha_M^N(A)) - \rho_M(A)| \leq 2\delta. \quad (2.9)$$

Letting $\delta \downarrow 0$ proves (1.4).

⟨1⟩ 5 Prove projective consistency.

Fix $M \leq M'$ and $A \in \mathcal{W}_M$. For every $N \geq M'$,

$$\alpha_{M'}^N(\alpha_M^{M'}(A)) = \alpha_M^N(A). \quad (2.10)$$

Apply ω_N , take $N \rightarrow \infty$, and use (1.4) at scales M, M' :

$$\rho_{M'}(\alpha_M^{M'}(A)) = \rho_M(A). \quad (2.11)$$

This is (1.5).

⟨1⟩ 6 Construct the inductive-limit state.

On the algebraic union prescribe

$$\Omega_0(\alpha_M^\infty(A)) := \rho_M(A). \quad (2.12)$$

Equation (2.11) makes this well-defined. It is positive, unital, and bounded by one on each embedded algebra. It therefore extends uniquely to the norm completion $\mathcal{W}_\infty$, proving (1.6). $\square$

3 Exact continuum-identification criterion

Corollary 3.1 (identification on one common algebra). Let ω_{ct} be a state on a continuum Weyl algebra $\mathcal{W}_{\text{ct}}$, and let

$$\beta : \mathcal{W}_\infty \longrightarrow \mathcal{W}_{\text{ct}} \quad (3.1)$$

be an injective $*$ -homomorphism satisfying $\beta \alpha_M^\infty = \beta_M$. If

$$\chi_M(\xi) = \omega_{\text{ct}}(\beta_M(W_M(\xi))) \quad (3.2)$$

for every M, ξ , then

$$\Omega_\infty = \omega_{\text{ct}} \circ \beta \quad \text{on } \mathcal{W}_\infty. \quad (3.3)$$

Proof. Equation (3.2) and Theorem 1.1 give equality on every Weyl generator, hence on $\mathcal{D}_M$, hence by norm continuity on every $\mathcal{W}_M$. The union of the embedded $\mathcal{W}_M$ is norm dense in $\mathcal{W}_\infty$, so the equality extends to all of $\mathcal{W}_\infty$. $\square$

Remark 3.2 (the exact remaining interacting statement). For the finite-torus Wick-quartic states from the model sheet, the unresolved scalar statement is

$$\lim_{N \rightarrow \infty} \omega_{N,L,\lambda,1}(\alpha_M^N(W_M(\xi))) = \omega_{P(\phi)_2,L}(\beta_M(W_M(\xi))) \quad (3.4)$$

for every fixed M and ξ . Proving (3.4) closes the static, full-sequence, identified fixed-coarse theorem by Corollary 3.1. Neither Theorem 1.1 nor Corollary 3.1 asserts (3.4).

4 Compatible symmetry

Proposition 4.1. *Suppose $\vartheta_N \in \text{Aut}(\mathcal{W}_N)$ satisfy*

$$\vartheta_N \alpha_M^N = \alpha_M^N \vartheta_M, \quad \omega_N \vartheta_N = \omega_N. \quad (4.1)$$

Then every ρ_M and Ω_∞ constructed above is invariant under the induced symmetry.

Proof. For $A \in \mathcal{W}_M$,

$$\rho_M(\vartheta_M A) = \lim_N \omega_N(\alpha_M^N(\vartheta_M A)) = \lim_N \omega_N(\vartheta_N(\alpha_M^N A)) = \rho_M(A). \quad (4.2)$$

The inductive-limit assertion follows first on the algebraic union and then by norm continuity. $\square$

5 Why the cited anharmonic LR theorem does not cover the Wick quartic

Nachtergaele–Raz–Schlein–Sims, Theorem 4.1, assumes for the on-site perturbation V that

$$V \in C^1(\mathbb{R}), \quad V' \in L^1(\mathbb{R}), \quad \kappa_V := \int_{\mathbb{R}} |w| |\widehat{V'}(w)| dw < \infty. \quad (5.1)$$

Proposition 5.1 (exact hypothesis failure). *For every $\lambda > 0$ and $c \geq 0$, the Wick quartic*

$$V_c(u) := \lambda(u^4 - 6cu^2 + 3c^2) \quad (5.2)$$

does not satisfy $V'_c \in L^1(\mathbb{R})$. Hence the cited theorem cannot be applied to this interaction.

Proof in Lamport form

$\langle 1 \rangle 1$ Differentiate the stated polynomial.

$$V'_c(u) = 4\lambda u^3 - 12\lambda c u = 4\lambda u(u^2 - 3c). \quad (5.3)$$

$\langle 1 \rangle 2$ Obtain a lower bound on the tails.

If $|u| \geq \sqrt{6c}$, then

$$|u^2 - 3c| \geq \frac{1}{2}u^2, \quad (5.4)$$

and therefore

$$|V'_c(u)| \geq 2\lambda|u|^3. \quad (5.5)$$

⟨1⟩ 3 Integrate.

For every $R \geq \sqrt{6c}$,

$$\int_R^\infty |V'_c(u)| du \geq 2\lambda \int_R^\infty u^3 du = \infty. \quad (5.6)$$

Thus $V'_c \notin L^1(\mathbb{R})$, so the first analytic hypothesis in (5.1) already fails. $\square$

Remark 5.2. Proposition 5.1 excludes only this citation route. It is not a nonexistence theorem for quartic Lieb–Robinson bounds. A replacement would require a polynomial-specific energy-weighted argument, with constants controlled uniformly in the resolution.

6 Revised proof obligations

Item	Status	Exact content
Projective cofinal-subnet existence	PASS	Proved independently in the compactness companion.
Full-sequence reduction to Weyl generators	PASS	Theorem 1.1.
Continuum identification once generator limits are known	PASS	Corollary 3.1.
Interacting characteristic limit (3.4)	FAIL	This is the remaining static finite-torus theorem. It requires a uniform Euclidean or cluster-expansion comparison.
N -uniform gap and local second moment	FAIL	Not needed for the static weak-star criterion, but still needed by the quantitative LPPL route.
Quartic interacting LR estimate	FAIL	The cited theorem assumes (5.1), which the Wick quartic violates by Proposition 5.1.
Local-state-norm rate	FAIL	Pointwise convergence on a norm-dense set does not imply uniform convergence over the norm-one ball.

References

- [1] B. Nachtergaele, H. Raz, B. Schlein, R. Sims, *Lieb–Robinson bounds for harmonic and anharmonic lattice systems*, Commun. Math. Phys. **286** (2009), 1073–1098.